Yad Hanadiv × MindCET

Event Transcript

From System-Oriented to Learner-Oriented EdTech

Yad Hanadiv × MindCET shared-learning series · Session 1 of 6 · Jerusalem, 15.10.2026

About this transcript: edited from an automatic (ASR) transcript without speaker identification. Fillers and false starts were removed and fragments merged into paragraphs; passages the ASR garbled are marked [unclear] or summarized in brackets; words in [square brackets] are editorial reconstructions. Speaker attributions marked '?' and names in brackets need checking.

Contents

09:23–10:17 Pre-event setup

10:17–10:23 Opening remarks

10:23–10:34 Round of introductions

10:34–10:44 Transcription gap

10:44–11:29 Fireside chat with Josh Schwartz: lessons from 25 years of EdTech investment

11:30–11:46 Break

11:46–12:24 Fireside chat with Prof. Arnon Hershkovitz: data, educators and AI

12:24–12:56 Fireside chat with Prof. Sheizaf Rafaeli: networks, links and the connections between the dots

12:56–12:58 Closing

Pre-event setup

09:23–10:17

[Sound checks, screen setup and small talk before the meeting started. Omitted.]

Opening remarks

10:17–10:23

Main speakers: Moti Taubin (Director of Education, Yad Hanadiv) · Avi Warshavsky (MindCET, host)

Summary: Moti Taubin, Director of Education at Yad Hanadiv, opens the series: the shift from system-oriented to learner-oriented EdTech is a useful frame but not yet an answer. Before AI drives the next wave of investment, the field should understand what worked and what failed.

Moti Taubin (Director of Education, Yad Hanadiv)

10:17:01 Good morning, everyone. We're about to begin.

10:17:05 And please, out of respect to our guests who are going to speak next, we'll be in English most of the day. Nice to meet you, my name is Moti Taubin, I'm the Director of Education here at Yad Hanadiv. There's a famous joke in Israel about how long a conversation takes... It took us, I think, less than 10 seconds.

10:17:40 Good morning, everyone. Welcome to Yad Hanadiv. Avi, thank you for initiating this series and for bringing together such an experienced and varied group of colleagues and educators. The people in this room have spent years building educational technology, researching it, investing in it and trying to make it work. For this reason, I shall keep my opening remarks short.

10:18:14 The title of this series points to a move from system-oriented EdTech to learner-oriented educational technology. I think this is a useful perspective, but not yet an answer. Technology is becoming increasingly personal and individual. Learning, however, still depends on relationships: teachers, classrooms and public systems. This is something we would like to emphasize strongly, here and in the other meetings. The student sees technology as a tool, but the school also has to take into consideration teachers' daily schedules, curriculum, assessment, data privacy, procurement and accountability.

10:19:01 Education systems have a remarkable ability to absorb new technology without necessarily changing the way learning happens. Sometimes that stability is a strength; other times it is exactly the problem. This is why it is so important to begin with a retrospective. EdTech has promised personalization, better use of data, stronger motivation, wider access and more effective teaching. Some of these promises have been realized, others have not. Before AI drives the next wave of investment and experimentation, we should understand what failed, why, and which assumptions about students, teachers and schools proved too simple. Israel is very good at importing technology. We should try not to import the mistakes at the same speed.

10:20:07 At Yad Hanadiv, our starting point is clear: equal opportunity is the goal, technology is the means. Our education strategy focuses on language and mathematical literacy and on students' self-efficacy in primary schools. In middle school, we're also examining AI literacy and the ability of students to learn effectively with AI. We believe technology can accelerate school improvement, but only when it rests on strong foundations: capable leadership, [sound pedagogy?], skilled teachers and routines that keep the focus on learning.

10:20:48 The discussion has therefore already moved beyond whether AI should enter the classroom. We all know the answer: it is already there. The harder question is whom will it serve? Will it help students who need more support to practice, think and become more independent learners? Or will it mainly benefit students who already know how to learn, while enabling others to produce polished work without necessarily deepening their understanding?

10:21:18 This is also an important moment for Mindcet, as it enters a new chapter as an independent organization, and best wishes for the week you had, which all of us enjoyed. It carries both

responsibilities and opportunities to help the field distinguish between technology that is innovative and technology that genuinely improves learning. The quality and validation framework that Mindcet is developing can make an important contribution. It should help schools and systems ask simple but demanding questions: Does it work? For whom? Under what conditions? And at what cost?

10:22:04 Over these six meetings, I hope we move from curiosity about tools to discipline about outcomes; from pilots to the conditions required for scale; and from asking what AI can do to asking what schools and public systems must do differently. If we leave this process with fewer slogans, better questions and a clear view of what should be funded, tested, regulated, scaled or [stopped], then we can achieve meaningful results.

10:22:43 On a personal note, I really hope these serious meetings will make us all think, consult with each other and decide bravely what we want to complete and improve in the Israeli education system, and what should be cancelled and re-examined, tested really and honestly. So thank you, and I look forward to today.

Round of introductions

10:23–10:34

Main speakers: Avi Warshavsky (MindCET, host) · Dr. Ilan Ben Yaakov (Mindcet Labs) · Dr. Cecilia Waismann (Mindcet Labs)

Summary: A round of introductions: entrepreneurs, investors, researchers and people from the Ministry of Education, the National Library, the Davidson Institute, Yad Hanadiv and Mindcet. Many names were garbled by the ASR. Avi closes with the story of the tool Ilan built for the conference.

[Names below are as heard in the automatic transcript and need checking. Unclear passages are omitted.]

Avi Warshavsky (MindCET, host)

10:23:15 Avi, thank you very much. It's been a lot of fun to see all of you here. Maybe we'll start with a quick introduction, and then we'll dive in.

From the audience

10:23:40 Hi, I'm Zvi [Kanish], founder of TeachMe, a platform for teachers to learn from their own challenges.

10:24:03 [Dr. Noa Sher:] I'm a researcher and consultant in the field of AI and education over the past three or four years, and before that in technology, initially working mainly for Educate Ventures Research and UK-based companies, and now also as an academic advisor for [an institute of applied AI in education].

Josh Schwartz (East Wind Advisors)

10:24:55 Good morning, I'm Josh Schwartz, founder and managing partner of an investment bank based in New York City called East Wind Advisors. For 25 years we've been active in the education and EdTech area.

From the audience

10:25:30 Eyal [Rav-Noy], co-founder of Capit Learning. We have a literacy platform.

10:25:48 I'm [Roy Tzikurel, "Tziko"], the CEO of AI-IL, the national initiative for AI literacy.

10:25:59 I'm [Dr.] Yossi [Elran], head of innovation at the Davidson Institute of Science Education, the educational arm of the [Weizmann Institute].

10:26:26 Hi, I'm [Ido] Levita, head of research at [the Manor Center].

Prof. Arnon Hershkovitz (Tel Aviv University)

10:26:45 Hi, I'm Arnon Hershkovitz, from Tel Aviv University.

From the audience

10:26:57 Hi, I'm [name unclear]. [...] [Dudu Bleicher (Jotit)?:] One of our fields deals with preserving handwriting in the digital age.

10:28:24 Good morning, everyone. I'm Andrea [Nickel?], director of development and innovation here at [Yad Hanadiv?]. Happy to be here today.

10:28:33 Good morning, I'm Irit [Touitou], from CET. I've been [working with] Mindcet for the last 13 years, [...] now in an independent way.

10:29:14 I'm [Michal Kihan, Branco Weiss], head of the department for elementary [education]. We work intensely in elementary schools in social and geographical [peripheries], and we've been dipping our hands and feet into technology.

10:29:38 Hello, I'm Gili [Leibushor]. I'm the head of [the Ora] initiative, one of the [unclear] initiatives, and we work with local municipalities to cultivate [...].

10:29:59 Good morning, I'm [Neta Shapira], Head of Education at the National Library. We focus on humanities teachers across the country, including Orthodox teachers, and we are very busy thinking about how to support processes of building identity, sense of belonging and awareness of everyone's heritage in this new era. The humanities are at the core of this. I'm very thankful for the invitation, because I think the discussion should be deep: not just how to do things and how to use them in class, but why and what for.

10:31:02 Hi, [Liao?], EdTech [at the Ministry of] Education. [I] run all the EdTech projects and strategy here in Israel. Happy to be here, and thank you.

10:31:20 Hi, everyone, my name is [Dr.] Nir [Tsuk]. I teach innovation at NYU, New York University. We launched an institute eight years ago which tries to teach innovation slightly differently: we do

executive education, and we teach innovation as a mindset, a language, a culture. Usually universities teach entrepreneurship at the business school, for startups and high tech. We say that if it's a language, everybody should speak it. So the first course was for rabbis, then for social workers, sex trafficking survivors, medical doctors and so on, online and offline, for the past eight years.

Dr. Cecilia Waismann (Mindcet Labs)

10:32:13 I'm Cecilia Waismann, I think most of you know me as well. Thank you so much for being here. We have big hopes for this group.

Dr. Ilan Ben Yaakov (Mindcet Labs)

10:32:24 I'm Ilan Ben Yaakov, the VP of Learning Experience at Mindcet.

Avi Warshavsky (MindCET, host)

10:32:32 And Ilan is also the magician behind all the technology we are using today, and the whole week. Everything you saw on screen is thanks to him.

Dr. Ilan Ben Yaakov (Mindcet Labs)

10:32:48 [What is it? The software.] It's something I developed for the conference.

Avi Warshavsky (MindCET, host)

10:33:05 Don't worry, I think your question is actually connected to the thing we want to have in mind when thinking about EdTech in transition. Ilan developed this tool for the hackathon, and Josh, who was the main audience for this tool, said during the hackathon that it's terrible. So Ilan sat the night after the hackathon, which was also the night before Tuesday's conference, re-prompted everything, and it works much better now. I think it's a good symbol for this whole changing industry. I don't think you wrote any direct code? No. And the outcome is amazing, and we are all partners in the process. It's okay to say it doesn't work, then we tweak it and something else comes out.

Transcription gap

10:34–10:44

[No transcription was captured between 10:34 and 10:44. This probably covered the context and goals presentation by Avi Warshavsky (and Moti Taubin), which was (later referenced as showing Edison in the early 20th century) and the start of the conversation with Josh Schwartz.]

Fireside chat with Josh Schwartz: lessons from 25 years of EdTech investment

10:44–11:29

Main speakers: Josh Schwartz (East Wind Advisors) · Avi Warshavsky (MindCET, host)

Summary: Josh Schwartz, an investment banker who has worked in education for 25 years, talks about founders' blind spot (passion without market realism), ReUp Education and PowerSchool as success stories, the capital markets' overcorrection after the COVID bubble, the critique in The Digital Delusion, and what will not be replaced: creativity and inspiration. Includes an overrated/underrated round and audience questions, among them the Ministry of Education's 720 program.

Josh Schwartz (East Wind Advisors)

10:44:37 [...] make the world a better place. Fortunately, the industry has grown up right around the time I made that decision, and it's now a proper industry with a lot of investors and a lot of excitement. You could even say it's kind of sexy.

Avi Warshavsky (MindCET, host)

10:45:01 Yes, and I can testify, working with Josh for a pretty long time, that your perspective goes much beyond the investment bank, and I think it's extremely relevant for this room. I wonder if there is a single point you see, working with so many entrepreneurs, that is a blind spot for them.

Josh Schwartz (East Wind Advisors)

10:45:34 Perhaps one that I've seen is that entrepreneurs who work in this sector bring so much passion for it. The funny thing about the sector is that everybody's got a point of view, because everyone was educated, understands the shortcomings of their education, and often develops thoughts about how to solve the problems. But you can't solve problems in a vacuum without understanding who else is out there and what some of the challenges are, like gaining distribution, which is one of the hardest things, especially in K-12. So the blind spot is bringing so much passion, which is great, but not tempering it with the realities of the market. We often find companies starting off without understanding that this has been done before, maybe not as well as they think they'll do it. And the reality is that the successful companies we work with don't always have the best product. That's a shame, but that is the reality. It needs to be a great product, it needs to deliver outcomes and have a great value proposition, but the best product doesn't always win.

Avi Warshavsky (MindCET, host)

10:47:18 I can totally resonate with that from the Mindcet experience. It's always a dilemma whether you want to be the boomer that kills the idea on the spot, or assume they will figure it out as they go. I wonder if there is one company that captures the story of EdTech, to get a bigger picture of EdTech in the US.

Josh Schwartz (East Wind Advisors)

10:47:47 I could cite some very successful companies. We love talking about Duolingo, because that's really the poster child. But perhaps it's more interesting to talk about one or two companies that my firm has worked with, because I can speak more intimately. One in higher ed, from a few years ago, called ReUp Education. They were looking to solve the issue that completion in higher ed is so challenging. They dealt mostly not with the Ivies but with the 99% of other institutions serving normal students, who are perhaps working while trying to upskill and gain social and economic

mobility, and for whom it is challenging to complete a degree. That is hugely costly for universities, because they spend a lot to market, anticipate the tuition, and then find there's a failure along the way. Had they been able to intervene and keep the student on track, the student would complete, which is beneficial economically, and would have better skills to thrive in the economy.

10:49:46 So they built a platform used by coaches, with algorithms (this was years ago, not leveraging foundation models as we know them today) that enabled coaches to identify which students they could help. Because of the support from the technology, one coach could support thousands of students. It was a very smart economic model, a true partnership with the universities: no upfront license fees, but if they were able to re-enroll a student who had stopped out, they would get a piece of the tuition, which universities were happy to share. We were able to bring in a financial partner and more capital, and since that transaction they've thrived. They've gone from working with universities to working with US states, which have plans for economic development and know where there are labor shortages. The states use the company's platform to direct adult learners into these areas, get the education they need and join the workforce. It has become a strategic capability for states.

Avi Warshavsky (MindCET, host)

10:52:06 It's a great example, and very relevant to what we're speaking about, thinking about underserved populations. Moving to the general mood regarding EdTech in the US in the last three years or so, would you like to add some color?

Josh Schwartz (East Wind Advisors)

10:52:26 Sure. To sum it up in one word, from the capital markets perspective, it's just more discerning: more selective, more real. During COVID there were artificial drivers of growth, and they propped up a lot of companies that in ordinary times would not have thrived. A tremendous amount of stimulus money produced distorted results, and the capital markets, where there's plenty of sophistication, get fooled by false positives all the time. So we saw a tremendous amount of investment in 2020, 2021, even 2022 go into some of these companies. The failure was to assume that it was going to continue and not to understand that it was artificial, due to the stimulus. When the stimulus ended, it took a while to work its way out of the system, but once it did, a lot of these companies just crashed. That created a vicious cycle in the market: investors and buyers scrambling to figure out what to do with these companies, or just forgetting them and moving on.

10:54:15 My feeling is there's been an overcorrection, and there's a lot of negative sentiment, including that book Avi mentioned, which, as I said the other day, is being handed out at school board meetings. In my experience the pendulum always swings too far in one direction and then too far in the other. I'm a firm believer that the mistake we made in the first and second generations of introducing technology was to believe that technology is going to solve all of our problems, and solve them very quickly. This is a long game, and I think the same will apply to AI. We're still in charge. The teachers and instructors are still at the center of everything. At the end of the day this is a tool. I read so much about how an AI revolution is going to take over the classroom; we don't believe that, and I don't think we'll let it. It's an amazing tool, we'll figure out how to incorporate it into everything we

do, the tide will rise, and 10 or 15 years from now I think it'll be clear how we've benefited from it. We've got some really complicated decisions to make, but I'm confident we'll be able to.

Avi Warshavsky (MindCET, host)

10:55:48 And talking about the book, you read it.

Josh Schwartz (East Wind Advisors)

10:55:51 Yes, I read this book. Anybody else read it? [Horvath], yes. It was recommended to me by one of the founders of the [Shefa] School in New York, a great school mostly for kids with learning challenges like dyslexia, which has been an amazing model, but they're coming from a 'less technology' approach. I noticed a couple of things. One: on every page the author was criticizing EdTech, but he never said what he meant by EdTech. I gathered that what he really meant was Chromebooks and screen time. He failed to distinguish, lumping everything into EdTech, which became a boogeyman you fear but never quite see. I've worked with a lot of amazing EdTech companies generating incredible outcomes, and I felt it was a very superficial treatment. Two: correlation isn't causation. He focused on results that can occur because of school closures and too much time on smartphones. I didn't think it was a deep analysis of what didn't work.

10:57:43 That being said, it's a legitimate criticism, and it's good to have that challenge come to us in the ecosystem, because we definitely need to do a better job. But critics of EdTech need to understand that the label is insufficient to describe the power of introducing technology in a variety of ways. One of the really successful areas in the markets where I work is school safety, and I consider that EdTech. It doesn't have a direct impact on learning, but a safe school environment is table stakes for a good learning environment. These technologies introduced capabilities that didn't exist and that would never enter Dr. Horvath's mind. So there have been a lot of powerful examples that were missed.

Avi Warshavsky (MindCET, host)

10:59:04 Yes, it's The Digital Delusion, which came out relatively recently.

Josh Schwartz (East Wind Advisors)

10:59:32 [...] When I started working in this sector, in 2000-2001, there really wasn't much of a sector. There was some interesting innovation around online delivery in higher education. In K-12 there were some sleepy companies. One was a company called PowerSchool. It was acquired by Apple in 2001, believe it or not, for about $62 million in Apple stock. A few years later they got tired of it and sold it to Pearson, the big publisher. Pearson owned it for maybe 10 years and then sold it to a financial sponsor, Vista, for $350 million. PowerSchool is a student information system. It's like a dial tone for schools: you need to have it, and once you have it you never rip it out. But it was operating in a silo. There was all this incredible student data on the administrative end, and it didn't talk to the classroom.

11:01:08 This financial sponsor bought it for what seemed an absurd price, $350 million, and over time connected the back office to the classroom, so there was a more seamless flow of data between the

student information system and the classroom, feeding back into the system. They created something called Unified Classroom and added a number of capabilities, never getting to instructional materials, but including systems for teachers. We ended up taking it public, and then in 2024 it was acquired by another private equity firm for $5.6 billion. For me, the growth from $62 million to $5.6 billion is just a proxy for something they must have been doing right. Nobody can tell me that connecting all the data in the back end to the classroom doesn't have an important and potentially transformative effect, and it happened behind the scenes. Nobody really talks about it, but there are a number of things like that over these couple of decades that are important to know.

Avi Warshavsky (MindCET, host)

11:02:58 It's a great observation. Going back to investors in the field: I see lots of rhetoric among funds that they care a lot about evidence. Is this just rhetoric, or is it something real?

Josh Schwartz (East Wind Advisors)

11:03:18 I think the best investors do care about that. The dilettantes, and maybe this is one of the reasons they got fooled by false positives, were more focused on growth and usage and not on outcomes. But the best investors (an example is [A-Street?] and Marc Sternberg, who was supposed to come on this trip) are as focused on outcomes as anyone. The best investors care about outcomes, and they recognize you're never going to have a sustainable solution if it's not generating great outcomes. When there's a lot of hype, industries attract a lot of dilettantes evaluating the wrong things, but now, in a more discerning market, more investors are looking at outcomes.

11:04:26 The challenge, and the reason this is a long game, is that especially in K-12 getting distribution is really, really hard. It takes a lot of capital and time; you need to generate research and then distribution. Investors seeking return on investment don't have a lot of patience for that, and it's hard for them to fund it, so there need to be solutions that bridge the gap. There are great solutions and great companies that never had enough time and resources to get to the other side. And now the capital markets have really seized up, and it's harder for new companies to get funded. They need to be AI-native more often than not. Great solutions that are needed in the market have to weather a very challenging period in which they need to generate research, show efficacy and build out distribution.

Avi Warshavsky (MindCET, host)

11:06:06 Great. I think it's a very interesting positioning in terms of how we divide the labor between more impact-oriented resources and more business-oriented ones. I have many more questions, but we should open the floor soon. Before we do: I really enjoyed the underrated/overrated session on Tuesday, so I want to do it again. AI tutors?

Josh Schwartz (East Wind Advisors)

11:06:47 I would say overrated, if you believe they're going to operate as more than a support to extend [the teacher]. There's so much hype around it. It's challenging, particularly with younger students, with privacy, and with the hostility now towards [kids] in front of screens.

Avi Warshavsky (MindCET, host)

11:07:24 Adaptive learning?

Josh Schwartz (East Wind Advisors)

11:07:33 I hate to give both answers to each question, but adaptive learning was the hottest thing 10 years ago, and it turns out a lot of the companies that claimed to be adaptive really weren't. So definitely overrated from 10 years ago. But it has morphed into personalization, represented by the promise of AI, and there it's probably underrated, in the sense that true, effective personalization is going to be incredibly powerful for differentiating between learners. We talked about models like Alpha School, and I'm somewhat negative about its ability to disrupt. What I hate about adaptive learning is the idea of kids sitting in front of screens in isolation, not working together, which is probably more like what they'll experience in the work world. So true personalization is exciting, but it has to be designed so you don't sacrifice everything else: social bonding and collaboration.

Avi Warshavsky (MindCET, host)

11:09:23 Khan Academy?

Josh Schwartz (East Wind Advisors)

11:09:27 I'll say rated about right. It was the hottest thing since sliced bread when it came out. It's interesting that they've accomplished so much as a nonprofit, and they were a first mover with AI, creating something called Khanmigo, with a failure to thrive. Sal Khan, the founder, has been very honest about where they failed. To sum it up, it's a lack of [engagement]: you're betting on a kid being able to use it at the right time, when they feel they have a need. An Israeli-founded company has a different pedagogical approach, where everything is aligned to the curriculum, so it will be really interesting. So, rated about right: kudos for being a first mover and solving an important problem, along with some sober perspective.

Avi Warshavsky (MindCET, host)

11:10:55 Thank you. Maybe we'll now open the floor for questions, about 15 minutes. I'm sure we'll have many more than that; the ones we can't cover now, we'll find a way to address later. We also have a Mentimeter, so in the next sessions you can also post questions online. I saw [Sheizaf] raising his hand.

Prof. Sheizaf Rafaeli (Shenkar)

11:11:27 Thanks for bringing up Khan Academy, because that's an introduction to my question: nonprofit versus for-profit. [...] Musk versus OpenAI [...]. I'd like your take on where education should go: should it be [free / nonprofit] over an extended period of time?

Josh Schwartz (East Wind Advisors)

11:12:36 At the end of the day somebody's paying for it. Public schools are free, but someone funds them, and funding has skyrocketed in the last 10 years. Do you know what the per-capita funding is

in New York public schools? I think it's about $50,000 per student, more than other states, and the results are dropping. So it's not a funding problem; it's something else, fundamentally. I don't know if it could ever truly be free; if it is, it's free in the sense that it's publicly funded. It's sort of like health care: if you join the HMO you're confined to a certain set of doctors and services, and if you pay a bit more you expand it. It's unfortunate, because it ends up increasing the divide between the haves and the have-nots.

11:14:46 Some investors are funding alternatives to school, and some can probably figure out models that are also free, but you have to get somebody to pay. One interesting example from K-12, which worked for a while until it didn't, is a company called EverFi. They had the insight that most of K-12 is literacy and math, then science, and social studies is sort of an afterthought. But what about financial literacy, anti-bullying, digital literacy and other subjects that aren't part of the curriculum? There's no money in the school system to pay for them, so they got sponsors, and it was free to the district. I think that's a really interesting model, especially in higher ed, where successful schools in the future will have to work on the connection between college and career. Employers spend so much on upskilling, retraining and retention that maybe they'll pay colleges to produce graduates with the skills they actually need. There are smart ways to reconstruct some of the models and bring down costs, and technology should help, but I don't see how we'll ever get away from the phenomenon of private options with a lot of money. Although there's also a lot of pressure on them to provide scholarships, so there's not as much inequity between public and private.

From the audience

11:17:26 [Partly unintelligible. A participant asks about the Ministry of Education's 720 program and mentions that their organization has just launched with dozens of schools.]

Josh Schwartz (East Wind Advisors)

11:18:54 My homework is to learn more about 720. Can you explain what 720 is? I somehow assumed this is something we all know.

From the audience (Yad Hanadiv)

11:19:16 It started last year with robotics and science. Last year it was in 32 schools; this year it's 140 schools. It's the second year of the pilot, and the year after that is supposed to be full deployment for all schools, according to the Ministry of Education's plan. It offers an environment where students learn independently, accompanied by AI tools that adjust the flow and the patterns of learning. Teachers take a totally new role: they see the class as a whole, who advances faster and who is slower. The system doesn't really include teacher-led instruction or group learning; it focuses on the self. Theoretically, I understand the next stages will have to [add that].

From the audience

11:20:56 Does it replace the traditional lessons in these subjects?

From the audience (Yad Hanadiv)

11:21:02 Yes, it replaces the traditional [lessons]. This year it's 140 schools, but it's going to be all the schools in Israel, starting with [six] subjects. In the vision, eventually, yes, it's supposed to replace them.

Josh Schwartz (East Wind Advisors)

11:21:57 People like to say it's the difference between the sage on the stage and the guide on the side, and this sounds more like a guide on the side. Is it aligned to the curriculum?

From the audience (Yad Hanadiv)

11:22:24 Yes. The material is not generated by AI itself; it comes from curriculum suppliers. So the AI is really more like a wrapper. And the curriculum is centralized, not localized to the school.

From the audience

11:22:50 Can you say a word about the pilot results?

From the audience (Yad Hanadiv)

11:22:55 Last year there were small pilots. The results are not that [clear yet]. We visited, for example, an all-digital school, not a regular school in Israel, with a strong digital base. We talked with students and teachers, and they were very happy. Students advanced a lot more than usual, and teachers felt they reached more students, because they were free to move between students, to whoever needed them. That was very good feedback. In other schools the feedback was not [as good], partly for technical reasons such as Wi-Fi that didn't work. We hope our Yad Hanadiv board will approve a grant for the next stage, in order to measure and understand the effects of 720, together with the Ministry of Education's chief scientist.

From the audience

11:24:38 I'd like to ask about a deep regret you have as an investor, with your long-term experience. And another question: relating to what Avi showed about Edison in the early 20th century, what won't be replaced by EdTech, AI or any kind of technology? What won't change in our schools in the next generation of students?

Josh Schwartz (East Wind Advisors)

11:25:22 The second one first. [The teacher remains] at the core of everything. And perhaps it's recency bias, but I'm in the middle of a book by Rick Rubin, called I think [The Creative Act:] A Way of Being. He's a very successful music producer who has worked with everyone from Johnny Cash to the Beastie Boys. The book is about the act of creation in general: where inspiration comes from, where creativity comes from, and what you do with it. That's not going to be replaced. Good luck utilizing AI to manufacture inspiration; it's never going to come from technology. He writes a lot about becoming more aware; it's truly a way of being. Learners are going to need that, and we'll be well served to find ways to separate learners from technology when it comes to developing creative skills. That applies everywhere, in the arts but also in business.

11:27:45 At breakfast yesterday, somebody said he asked AI apps, 'What do you know about me?' I ended up doing that this morning, and it's shocking. Not only the facts, but what it derives from them about your personality. It's incredibly accurate. There's an interesting use case there, but we need to use it to replicate the things that are at our core and part of our spirit.

11:28:33 The regret is that there's just not enough [patient] capital for this, and in some cases capital has given this sector a bad name. But as Avi has said, we need to fail; we're never going to be perfect. Even though there have been failures and negative sentiment, the most successful entrepreneurs, tech moguls and investment funds can't resist education. It goes back to what I said before: everybody's got a point of view on education. When people have been successful in their careers and want to figure out what to do next, they all want to come back and transform education and correct the problems. It's never going to end, and it's great. You want to learn from those mistakes, but I am incredibly optimistic.

Avi Warshavsky (MindCET, host)

11:29:45 Thank you very much, Josh. Maybe we'll take a 10-minute break.

Break

11:30–11:46

[Break. Background conversations omitted.]

Fireside chat with Prof. Arnon Hershkovitz: data, educators and AI

11:46–12:24

Main speakers: Prof. Arnon Hershkovitz (Tel Aviv University) · Avi Warshavsky (MindCET, host)

Summary: Prof. Arnon Hershkovitz, an educational data mining and learning analytics researcher, explains that teachers barely use data systems: their decisions are intuitive and the systems feel like a burden. The discussion moves to how much a school should know about students' lives outside class, the difference between macro-level data and the personal relationship, and the risk that AI conclusions replace teachers' judgment. Arnon closes: we decide the goal of each lesson, and we need to teach both with AI and about AI.

Avi Warshavsky (MindCET, host)

11:46:03 Moti, I would like to invite you in. We are honored to have here Professor Arnon Hershkovitz. Maybe we'll start with a little introduction of yourself. Please scan the QR code: those of you who want to ask questions can put them in the Mentimeter as well, so even if we miss some of them face to face, we can go back to them.

Prof. Arnon Hershkovitz (Tel Aviv University)

11:47:08 My background is in mathematics and computer science: a BA in computer science and a master's in applied mathematics. I worked in industry as a programmer and algorithm engineer. Because of my background I was attracted to a new avenue of research, educational data mining. Back then it was quite new: the international community was just gathering around the notion that we can use data from digital learning environments to understand learning. Since then the terms have changed a bit; today learning analytics is the more common term, and we understood that it's not just about data, it's also about learners, context and processes. I shifted from a relatively narrow look at data from educational systems to how education stakeholders refer to data. So today I'm interested in data in education both methodologically, to understand learning processes, and for stakeholders, to help them understand what's happening.

Avi Warshavsky (MindCET, host)

11:49:46 Good. So maybe we start with the stakeholders, with educators and their way of interacting with big data.

Prof. Arnon Hershkovitz (Tel Aviv University)

11:50:02 Let's start with a small example, connecting to PowerSchool. Here in Israel we have [Mashov] and similar school systems, and in many cases they're still disconnected. When we studied how teachers and educators use these systems to better their teaching, we found that basically they don't. The reason is that what they get out is limited, while on the other hand they are required to put all kinds of things in, so it's a burden. Teachers' data comes from observations, conversations and of course grades. For many decades we've seen that teachers' data-driven decision-making is not really data-driven; it's mostly intuitive. Data could help them, and many of them want it.

11:51:52 We ran a study in the US, at university level, and asked instructors and counselors what kind of data they would like to have about their students to understand their progress. They wished for almost everything: starting from the obvious, grades of course, and going to sleep habits, participation in varsity teams, how students plan their learning, how they manage their day, whether they have hobbies. Then we ran similar studies with school principals, not in Israel, and again saw a diversity of data they want: how do students get to school, does someone drive them, do they take the bus, why are they late, everything social-emotional, the situation at home and who helps them. I'm trying to imagine where this data can come from, and whether the data EdTech products produce is too narrow in that sense.

Avi Warshavsky (MindCET, host)

11:53:54 Thinking out loud about the poor usage of data by educators: is there a good example of a sector or someone in other domains who really uses data for decision-making? Or is it a problem of human beings and not of educators? I don't know if [Sheizaf] has some insights about that.

Prof. Sheizaf Rafaeli (Shenkar)

11:54:37 Teachers are people too. Some institutions are more data-driven, and others, even social work, sometimes are, but there are always limits on what kind of data they collect and how it's organized. The more it's organized in a way that suits the decisions you want to make, the better use

you can make of it. You don't want to collect data for the sake of collecting data, or to be overflooded for the sake of being overflooded. It only helps if somebody makes sense of it and there's a thinking process behind it. There are so many professors making a career out of cognitive biases; that whole field is about the ways we disregard data.

From the audience

11:55:52 [Partly unintelligible. A participant describes a Ministry of Education plan for one system that will hold everything that happens with students in school, and big data that helps students develop through adaptive learning. The dilemma: students say they waste time on screens, so materials for 10th grade were redeveloped in all disciplines with three modes, only one of them on screen and two through inquiry-based learning. The ministry wants each student's AI agent to lead and help them; that's nice, but it can be dangerous, because students use AI [to avoid the work].]

Prof. Arnon Hershkovitz (Tel Aviv University)

11:58:33 We'll reach those questions soon, but let me hop in on this. We don't want teachers to be clerks, data-driven clerks. Their intuition, based on experience, knowledge and professionalism, is very important. So there's a trade-off. As a researcher I wish to get as much data as possible, but we should balance somehow; we don't want everything to be in front of the screen, of course not. Additionally, data-driven decision-making requires data literacy from teachers, which in many cases is missing. Just watching a dashboard doesn't mean teachers know how to accept this data and respond to it.

12:00:18 One study we ran here in Israel: when data was available to teachers in mathematics education, we compared what the machine sees and what the teacher sees, and how each reacts to the data and takes action. Spoiler: the findings were quite obvious. Each actor worked a bit differently, in a way that working together would have been [better] for both. The teacher had knowledge about her classroom: who the kids are, what they know, what she taught in previous lessons, what she wants them to achieve, the curriculum. The machine knew exactly what they did in the system, for how long, where they got stuck. When we personalized the content, either by the teacher or by the machine, student performance was about the same, because each took different decisions based on different data. So it's not that one is better; they should work together.

From the audience (Dr. Cecilia Waismann?)

12:02:00 I would like to say one thing. When big data became a big thing, there was a slogan: because we can doesn't mean we should. It comes back to me now. It always bothers me tremendously when we start justifying our failure with students by not having enough data to understand them, and then we want to know how they slept, how they ate, how they came to school. Throughout our years at Mindcet we've seen many extremely well-intended schools and companies that wanted to enter social networks to enhance students' social-emotional response. But should we be doing that? Is that the role of school, the role of educational solutions? There's a reason psychology explains that kids perform differently; kids are different when they leave their parents and go to school, and it's important for their development. So my question is: because today we can get data about everything, should schools be entering there? Would that enhance the educational solution?

Prof. Arnon Hershkovitz (Tel Aviv University)

12:03:25 I think there are two questions here. The first: should schools treat students as whole persons, with their social-emotional needs, their academic background, their activity on social media and so on? And the second, if our answer to the first is yes: how do we achieve this? Did I get you correctly?

From the audience (Dr. Cecilia Waismann?)

12:03:58 Yes. Of course education should see the human being as a whole. But should we invade the student when they are not at school?

Prof. Arnon Hershkovitz (Tel Aviv University)

12:04:05 That's a good question. If we agree that teachers and schools should look at students as whole persons, they should somehow get insights (I won't use the term data) into the students' lives. I'm not into connecting smartwatches and Facebook accounts.

From the audience

12:04:39 The first thing that comes to my mind: in order to see a kid as a person, as a human being, you don't need to know everything about him. If my kids' school were integrating [data on] what she does after school, I would be very [uncomfortable].

12:05:06 And I'll add that there's a big gap between caring about the child's welfare and collecting so much data. Maybe you shouldn't know that he slept with four [siblings] in his room. Maybe not knowing will help: the kid gets an opportunity to present himself as he wants to be.

Prof. Arnon Hershkovitz (Tel Aviv University)

12:05:47 We already sort of answered this question, because we had the unfortunate opportunity during COVID to bring teachers into kids' personal spaces. For some kids it was horrible and for some it was great; for some teachers it was great and for some it was a horror. But we were there already, and kids do separate and want to keep the separation between school and out of school. We've seen this with technology too. Back in the Facebook days there was a big question whether we should connect students and teachers via Facebook, and the Ministry published guidelines. What we heard from students was: Facebook is not for school, it's for our personal lives, so keep teachers apart. It also relates to bringing technology students know from their personal lives into school: some educators say, they use Instagram or ChatGPT, so let's bring it into classrooms. In many cases it creates resistance among students, because they need this separation. Don't get me wrong, I wasn't in favor of connecting smartwatches and cell phones, or doing what they do in China, with face-recognition cameras in schools where the teacher knows everything.

From the audience

12:07:53 I think this brings up the [micro] level, which is the relationship. There's no need to collect data there, because the teacher [knows the child]. Data is more for the macro level, and that's a different conversation: you want to be able to decide at the macro level whether, for example, kids

who are driven to school need more time in general, or whether to start the school day later nationally. That's a very different perspective from a specific child being late to school and the context of that child. There has to be a differentiation between these two levels. The teacher is supposed to be able to communicate with the number of students they have; if they can't, that means they have too many students, not that you're not collecting data efficiently enough. And to make macro-level decisions you don't need specific personal data, you need guardrails on what data you collect.

12:09:08 In more European contexts, these conversations start with privacy and guardrails: what does it mean to collect that much [government] data about children? Here it seems it's gone a bit too far, so that it's not even part of the conversation. It has to be: why is the data collected, who collects it, who stores it, who has access to it, how long is it stored, is it consensual, do parents agree? Again, start with the why. Some things you can get just by speaking.

Prof. Arnon Hershkovitz (Tel Aviv University)

12:10:16 Smartwatches aside: you mentioned data at the macro level, and I think it's crucial to have data at the micro level too. For the matter of this discussion, let's focus only on digital learning. When data is stored, managed and used wisely and effectively, it can help the teacher in the classroom in real time. Imagine a session where the class works on software in math or language. Giving relevant insights to the teacher in real time may be very important: things happening within the classroom, not the students' social and personal lives. That's something completely different. Everything happening outside the classroom, and any data that may be used later on, is a whole different question.

From the audience (Yuval Shraibman?)

12:11:44 I'd like to connect the discussion to something slightly different. Yesterday I talked with someone from a startup that uses AI very heavily. They got a complaint from a customer about degraded performance. The AI investigated all the events, also looked at the company's Slack for similar problems that had been reported, found the diagnosis, and had a response to the customer ready. I think this is the classroom. It's not about whether we collect the data or not: very soon the AI will come up with a diagnosis, a result and an instruction to the teacher. The risk is that we might lose the teacher's intuition, the human factor of dealing with a student with an issue, or there's potential for huge success on the other side. It's not only about data exposure and whether it reaches personal time; it's about the freedom and ability to employ judgment, and whether that still has the same place when the AI gets stronger and comes up with conclusions and selections, trying to level the playing field between teachers. It moves so quickly. It's not about exposure or even data collection anymore; it's about the authority of those conclusions and decisions. And it's three or four years ahead, not 20.

Avi Warshavsky (MindCET, host)

12:14:11 I want to follow up on what [Yuval] was saying: your take on the impact of the AI revolution on these kinds of systems. The technology is almost there, but that doesn't mean...

Prof. Arnon Hershkovitz (Tel Aviv University)

12:14:42 In the context of AI in general, including traditional AI from the very early days, there are three levels of intervention: descriptive, then predictive, then prescriptive. You refer to the third stage, where the machine magically knows what to do and tells the teacher. I think it's for us, the stakeholders, to decide whether we want that.

From the audience

12:15:47 Teachers are using it and students are using it already. If I'm starting a lesson now, I decide whether I plan it or let the machine plan it, and in the classroom I decide whether I allow AI use or not. It's complicated: on the one hand there are teachers who aren't so good at it and have no chance of getting good at it by themselves, and it's tricky because it's an easy solution. I think very soon people won't be able to do without AI. They're getting so used to consulting AI on every problem, personal or professional. The younger ones, and in two years maybe even us, won't ask themselves whether to use AI to prepare the lesson.

12:17:20 But there are consequences for what you decide. There are things AI does that we want and things we don't; it channels you.

12:17:39 I think it's a good question who the 'we' is, and what about the student element: do they have choices? My son is in 10th grade at a very high academic school in the States. I was talking to him yesterday and he said, 'I have this teacher who is crazy about AI.' I said, 'That's great for you, no?' He said, 'Well, I don't know. She gave us a study guide for this test, and it was all AI-written. Then I realized my test is going to be AI-written.' So he took the test, and he said, 'Mom, it was exactly [the study guide]. I only got three questions wrong.' And he didn't learn anything; he didn't internalize it. And it wasn't his choice.

12:18:27 That's why they usually have to balance between AI agents and agency.

Avi Warshavsky (MindCET, host)

12:18:38 I think that's a great framing for one of our next sessions, because it captures something: the old EdTech was something we controlled totally from the decision-maker side; the new EdTech happens out of our control. The framing of agency and agents is a great one to explore further. I'd like to go to Tzipi's comment: we talked a lot about educators as stakeholders, but not enough about students. Maybe this will be our last question. Arnon, your take on the way students work with these kinds of systems?

Prof. Arnon Hershkovitz (Tel Aviv University)

12:19:31 When I said 'we', I still believe that we do manage. It's for us to decide what our goal is: the goal of this meeting, of this class. When I set the goal, I decide how, where in the process technology is relevant and where it isn't, and what I tell my students they can or can't do. [...] As for students: we know, at least from higher education, that they use AI extensively, around 95% in some studies, and for many types of things. We mapped the ways education students use generative AI, and it's all over. But that doesn't mean they use it effectively, wisely or for their own benefit.

12:21:21 We as educators, as human society, are in charge of preparing the next generation to live, learn and work in an environment that is [full of] AI. It has been our responsibility since before generative AI. So we should teach them how to use technology effectively, and it requires us to engage better with technology, AI included. It's a dual perspective: teaching with AI and teaching about AI, and both are important because they are going to be living, learning and working in this world. It's very complicated: there's digital literacy, self-regulated learning, motivation, access, many issues, and we have 40 minutes, not 40 hours. We already know cognitive offloading is a big issue. So maybe we should control it somehow. Maybe it's okay to say: the goal of this lesson is to go back to writing, so in this session you write with no software and no computers at all. And in another session the goal is critical thinking about the outputs of generative AI, so of course we'll use it, but we'll reflect on it. I think this is the main way.

Avi Warshavsky (MindCET, host)

12:24:10 I think we should conclude. We'll thank Arnon. We have a lot to add, as in the session with Josh, and we'll try to follow up on some of the Mentimeter questions later on. Thank you very much, everyone.

Fireside chat with Prof. Sheizaf Rafaeli: networks, links and the connections between the dots

12:24–12:56

Main speakers: Prof. Sheizaf Rafaeli (Shenkar) · Avi Warshavsky (MindCET, host)

Summary: Prof. Sheizaf Rafaeli shifts the spotlight to networks: knowledge lies in the connections between the dots, not the dots themselves. He tells of his path from radio and TV to computer-mediated communication, Ted Nelson's Computer Lib/Dream Machines, the National Library steering-committee debate, and his claim that the most superfluous part of a library is the bindings. Education should teach network science and assess students by the links they create. A participant presents research on links in Stack Exchange forums and warns that AI flattens knowledge and makes the links disappear.

Avi Warshavsky (MindCET, host)

12:24:34 And Sheizaf, happy to invite you.

Prof. Sheizaf Rafaeli (Shenkar)

12:24:39 Do you need me here to ask you questions, or are you going to run through it alone? / You're welcome to ask me whatever you want, and I'll say whatever I want.

12:24:55 What I'll talk about is an attempt to put a spotlight on something that hasn't been mentioned yet: the issue of networks. I'd like to argue that networks are much more important than we accord them, that they are an incredibly important tool and substructure for much of what EdTech could achieve, and often an explanation for when and where EdTech [fails].

12:25:56 I thought, Avi, that you would start with me as you started with the others, with a request for some biographical information. So I'll give some. If, with Josh, we discussed the digital delusion, I'm a digitally deluded person. If, with Arnon, you spoke about data, then I'm a data fetishist: I believe in data and think we should work on collecting it. So if you want to pigeonhole me, I'm in the digitally deluded and data fetishist pigeonhole.

12:26:36 I started my career after my military service, where I worked with underprivileged children and in high schools, and came out very frustrated. Specifically with juvenile delinquents, it ended up with almost every single child I worked with going back to crime. So I started looking for solutions beyond the personal connection. This was back in the 1970s, and I became fascinated with the technology of the time, which was radio: you reach many more people, radio multiplies messages. I got involved in radio and later television production, including educational uses, with the Children's Television Workshop in the States, and I worked with [Radio Free Europe?], then the largest radio station in the world, and broadcast many hundreds of hours. But then I realized that radio and television don't know much about their audience; they measure ratings, but that's about it.

12:28:38 Then, while doing my PhD on the West Coast of the United States in the late '70s and early '80s, computers that large groups of people could use came into being, and I became interested in computer-mediated communication: new forms of communication that are not just two-way but collect data on all sides, convey messages and create emotions. When I became an entrepreneur I founded a few startups with technology intended to teach. Back in 1990, with a group of my students, we set up one of the first dotcoms in Israel, [Comda?], and one of the things we did was set up the first online Hebrew-teaching system.

12:29:55 By then I'd been in the business of teaching others for over a decade, and realized maybe I should read others. One of the books that made the strongest impression on me was Ted Nelson's Computer Lib / Dream Machines. If you haven't read it, I strongly recommend it. It was self-published. Ted Nelson is the person who, among other things, coined the word 'hypertext'. It's an eye-opener, written before there was an internet, about the idea of thinking about the links between things, not just the things. It's actually two books: Computer Lib starts at one end, and you flip it over and read Dream Machines from the other side. Creative, as most of the things he did. The idea of hyperlinks is what I'd like to focus on.

12:31:39 Several years later, one of the first groups that later evolved into a steering committee was put together to think about the new National Library. To oversimplify, the question posed to us was: should the new National Library invest the large sum that was planned (at the time $40 million, which sounded like a lot; it ended up costing roughly ten times that) mostly in the building, mostly in collections, or more in digital? The committee was composed of three groups: those advocating for building, those advocating for collections, and those advocating for digitization. You can guess which group I belonged to. And you know the result: a breathtaking building, a really big collection, and a lot of digitizing, but a lot less than we envisioned should have happened. I made the argument that they shouldn't build anything, but rather make everything available to everyone.

From the audience

12:33:59 What do you think about that today?

Prof. Sheizaf Rafaeli (Shenkar)

12:34:04 I still believe it. I did not change my mind. The building is fabulous and magnificent, the collections are great, and the National Library's activities are among the best, but I'm emboldened in the position I had then, which really was a losing position; I was made fun of. In the context of that steering committee I raised a question I'd like to pose to you, following Moti's question: if we were to redesign the library, or an educational institution, from scratch, under budgetary constraints, what would we do away with first? Let me warn you in advance, the answer is going to sound demagogic. I've raised this question many dozens of times with many groups. Some people say we should do away with librarians, because their only function is to shush us, or do away with the buildings. I'd like to propose that the most superfluous part of the library is [...] I'll pause for a second to let you fill in the blank.

From the audience

12:35:53 No, I think books are very important. I think they should be digital, but...

Prof. Sheizaf Rafaeli (Shenkar)

12:36:06 The bindings. They only get in the way. We would have a much better library if everything in the books were connected rather than separate. That's the basic idea: the book is broken; it's about connection. We've all grown up at different times (I'm over 70, some people here are half my age), but we've all been trained to include references, a bibliography, lists of sources, and we've all really been trained to disregard them, because most of us do it as a ritual, not for real. How many of you actually went and checked the references in a paper you wrote? Ted Nelson's point, and mine, the whole idea of hypertext, is that we should change that: the real essence of knowledge lies in the connections between the dots rather than in the dots themselves.

From the audience

12:37:26 Don't you need a basic understanding of [the field] in order to connect?

Prof. Sheizaf Rafaeli (Shenkar)

12:37:28 I'm getting to that. But first let's acknowledge that the connections between the dots are an important, maybe the most important, blind spot. That's also the opportunity of things being computerized: going horizontally rather than vertically, seeing items within the context in which they're located. It's not my idea; it has evolved over the last few decades under all sorts of titles: mind maps, prismatic thinking, hyperlinks. David Weinberger calls it 'Everything Is Miscellaneous': because it is digital, we can have things in different orders for different purposes, people and contexts, and that's their beauty. Miscellaneous organization is not disorder; it is the order, it is the power. To paraphrase Weinberger: think about your closet. Winter clothes are set apart from summer clothes, bright clothing from dark clothing, but there's always at least one drawer for things that can't be categorized. He says that in the digital world everything is uncategorizable, so let's respect and harness that rather than work around it.

12:39:21 Let's make use of it. Because it is digital, that's where we should strive to go, rather than run to an imposed order, which is a waste, cognitively, intellectually and educationally. We would do best if we honored miscellaneous organization. In Judaism it's called 'ein mukdam u'meuchar ba'Torah' (there is no earlier and later in the Torah): let's have everything acknowledged together and allow different people to create their own paths. Think about [the Talmud page]: it's hypertext. So it's not something invented by [computers], but it is something that can be enabled by digital, and too often it's not. Too often we still impose a linear structure rather than celebrating the opportunity.

12:40:41 There was a publication for the last 20-some years called Edge that I strongly recommend; it's all available online. It was a website and a series of books with an annual question, put together by [John] Brockman, and it is no longer being published. Every year, around October or November, he would come up with a provocative question and send it to two or three hundred of his closest friends around the world in different fields, asking for a single-page answer. One question was: what did you strongly believe in and no longer believe in? Another: what is the most important scientific idea of your age? Every year he would collect about 200 responses and publish a book. About 15 or 20 years ago I was in that group, and the question that year was: what is the scientific idea that is least known and that you think should be more widely known? I thought about it for about three seconds and answered: networks.

12:43:05 Because so much of our life is in networks, in things that are linked to each other. Biology is networks, brains are networks, businesses are networks, economies, oceans and so forth, and we spend too little on understanding networks and building technologies of networks. That was 20 years ago; our life is on the internet all the time now, and I think my argument is still true: networks have a technology of understanding and manipulating that we're still under-implementing. Teaching students to make connections, using technologies to encourage learning through drawing or creating links, and identifying these links is, I'd argue, more important than recognizing the nodes. A whole language of network analysis has evolved in the last 15 years and still hasn't assumed its rightful place. A basic statistics course is something almost everyone has taken and most have hated, and we still teach central tendency, standard deviation, correlation, the general linear model. We don't measure betweenness, we don't teach connectedness, we don't teach network math. We still teach the normal distribution as if it's the norm, and very little about power laws. We should be incorporating networks into technology and into teaching.

12:45:22 To bring this down from the conceptual level to the actual: in my own teaching over the last 20 years, and now as president, in a position of power, I've been encouraging, rewarding and urging the 500-some instructors I manage to teach by assigning the responsibility of mapping links. One way is to assign students to create Wikipedia articles, but assess them by the links, not the content: what the article is linked to, what other articles connect to it, who else linked to it and who reacted to it. In other words, focus on the network more than on the content. One of my collaborators, Carmel Kent, developed [a tool] for creating that, and other collaborators used this tool and other data to look at measures of links, and collaborative links, as a major measure of learning. Noa, if you want to say something about your work.

From the audience (Dr. Noa Sher?)

12:46:52 Yes. What I looked at was [Carmel's tool], and the other was Stack Exchange, the question-and-answer forums. You probably know Stack Overflow; it used to be a huge thing for developers to post questions and other people to answer them. Stack Exchange has forums about anything in the world: philosophy, biology, beer making. (By the way, these have dropped off dramatically now, obviously.) Users post questions and answers, and some include, within their answers, a reference to another question: 'this reminds me of something somebody else said', or 'this is exactly the opposite of something else', or 'from a broader perspective these could be related'. If you extract these links, you come up with a network, an invisible network: whoever uses these forums just sees questions, answers and maybe links, but not the bigger picture of how things relate.

12:48:34 To go back to what [Sheizaf] said about the book bindings: I don't think you should completely remove them, because then we become overflooded. We need organization, but what we need more is ad hoc organization that is dynamic by nature, and you can derive it bottom-up. Not everybody adds links, but some people do, and they become the latent organizers of the knowledge: if you take a step back, you see that things you didn't think were connected actually are, and you see structures within the network. And I think what's happening today with AI threatens to flatten some of what networks afford. With networks you can drill down to see how things connect, and go up to see how clusters form. Before, we had links, then search engines that organized things by links. Now AI takes the content, breaks it into very small bits and reconstructs it in a way that makes all the links go away. It simplifies, it reduces, and you get something reconstructed but without the depth it used to have, and often you can't reconstruct the links. In education and learning, I find it scary: people growing up in this world have never used the links. They don't go to the library and move from one thing to another; they get the flat information, and their way of thinking becomes flatter. You can go from depth to flattening, but you can't go back.

Avi Warshavsky (MindCET, host)

12:51:17 Can I just suggest, sorry, that we take this as an idea for next time: the oversimplification of information.

Prof. Sheizaf Rafaeli (Shenkar)

12:51:34 I share [her] concern. Clearly people are now turning to AI as their source of knowledge, not just in formal education but in daily life. We don't go to the elders, we don't go to Google, we don't go to Wikipedia; we go to AI with whatever question we have. One of the things we lose when we go to AI instead of Wikipedia is the links. This could be remedied if AI becomes better and realizes the importance of links. AI is networks, neural networks, but hidden networks rather than explicit ones. Our job in education is to make the networks explicit: make the measures explicit in assessment, make their creation explicit in assignments and requirements, and make networks themselves a subject of study. It's still the case that most higher-education graduates don't know enough about networks as a concept. They still think in terms of the general linear model, correlations, central tendencies and the normal distribution, and not about the way the world behaves, which is power distributions.

12:53:11 I'd recommend you look up my answer; I'll be happy to send it. It's a page-and-a-half argument about networks as an important idea, and because it's just a page and a half I resorted to

rhetoric: I said that the opposite of networking is not working, and that your network is your net worth. My current job is at Shenkar, an institution that focuses on engineering and design, and my job is to build a bridge that connects engineering and design. That's very difficult: designers detest engineers and engineers detest designers; neither side has any respect for the other. So I'm showing the designers that engineers have something to say, and the engineers that designers have something to say.

12:54:25 My last point: links are not just about the individual level and cognitive studies; they're also social. Creating links is something individuals do, but it's also social: if I link to your idea, if we collaborate in creating links, we learn a lot more than if we just create individual links. One of the first pieces of software I wrote and sold was a word processor that made it easier to create links. And it's only in the last three or four years, as I see in my grandchildren's schools, that teachers have started requiring students to create links. I'd argue that requiring students to create links is as important and useful as, maybe more than, requiring them to just take in content. If we equip students with tools that create links, we might have a much more networked system of learning.

Closing

12:56–12:58

Main speakers: Avi Warshavsky (MindCET, host)

Summary: Avi wraps up the day's three conversations and promises to send a summary and coordinate expectations before the next meeting. The series' first goal is building the group.

Avi Warshavsky (MindCET, host)

12:56:02 Thank you very much, Sheizaf. To conclude, first I'd like to out Lena [Ravad] and say that she wrote a PhD on this subject [...], and I think it's worth elaborating on it next time. Going back to the starting point today: the dialogue with Josh was about EdTech in general; with Arnon we dived mainly into decision-making based on data; and the way I see your talk, Sheizaf, is about the potential we're not using of internet culture. EdTech, traditionally, is a very linear, closed garden, as are most of the solutions we know. This was one of the hopes at the beginning of EdTech that was never fulfilled, and it's a very interesting point to look at right now, especially with the risk AI poses of taking it even further.

12:57:11 I'm sure there's a lot more to say and ask, but I'd like to respect the timeframe we designed. We'll send a summary of what we took from this day, and we'd be happy to have your input. Before the next encounter we'll try to coordinate expectations about the coming months. I know this was a good illustration of jumping around, of networks in a way, but I prefer it this way; I think it's much more fruitful than a linear structure. Please feel free to reach out to me and suggest ways to take the encounters further. We know the big picture, but within the plan it's really open for the room. And I'm sure I speak for Lena and Moti as well: creating this group is the first goal, and the rest will come. Thank you very much, I learned a lot today.